INCH-POUND

MIL-T-24940A
11 August 1993
SUPERSEDING
MIL-T-24940
30 September 1988

MILITARY SPECIFICATION

TROUSER, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a flame resistant structural firefighters trouser with a waterproof moisture barrier lining and a para-aramid/PBI blend outershell. In addition, the trouser has an aluminized para-aramid cover for proximity use.

1.2 Classification. The trouser covered by this specification shall be of the following types, sizes, and lengths (see 6.2):

1.2.1 Types.

Type I - Trouser, Firemen's, Structural

Type II - Trouser Cover, Firemen's, Aluminized Proximity

1.2.2 Sizes. 34, 36, 38, 40, 42, 44 and 46.

1.2.3 Lengths. Regular and long.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defence Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	- Bag, Plastic, Folded Garment
A-A-50195	- Thread, Aramid
ZZ-T-831	- Tubing; Rubber and Plastic
DDD-L-20	- Label; For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-E-20652	- Eyelet, Metallic, Rolled Flange Type, and Eyelet Washer
MIL-F-21840	- Fastener Tapes, Hook and Loop, Synthetic
MIL-C-24929	- Cloth, Laminated, Aluminized (Para-Aramid Base)
MIL-C-24939	- Cloth, Ripstop, Para-Aramid/PBI, Water Repellent
MIL-C-43774	- Cloth, Plain or Pajama Check Weave, Aramid
MIL-T-44100	- Thread, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1666	- Classification of Defects for Evaluating Quality of Aluminized Firemen's Clothing
MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirement Codes
MS-27980	- Fastener, Snap, Style 2

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standarization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 1971 - Standard on Protective Clothing for Structural Fire Fighting, 1991 Edition

NFPA 1976 - Standard on Protective Clothing for Proximity Fire Fighting, 1992 Edition

(Application for copies should be addressed to National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9990).

WARNING

THE TYPE I STRUCTURAL TROUSER AND/OR TYPE II PROXIMITY TROUSER COVER WERE DESIGNED TO MEET NFPA STANDARDS, AND MUST MEET THIRD PARTY CERTIFICATION TO ASSURE CONSTRUCTION, COMPOSITE MATERIALS, AND PERFORMANCE REQUIREMENTS MEET SPECIFIED REQUIREMENTS. THE CONTRACTOR SHOULD OBTAIN A COPY OF THE APPLICABLE NFPA STANDARD FOR CERTIFICATION PROCEDURES.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

JOINT COMMITTEE ON PRINTING (JCP) PUBLICATIONS

Government Paper Specification Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

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3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Basic material, type I (structural trousers). The basic material for the structural trouser shall be a water repellent para-aramid/PBI ripstop fabric conforming to MIL-C-24939.

3.3.2 Basic material, type II (proximity trouser cover). The basic material for the proximity trouser cover shall be of a highly reflective aluminum surface laminated to a knitted para-aramid base fabric conforming to type II of MIL-C-24929.

3.3.3 Moisture barrier. The moisture barrier material used for lining the structural trouser shall be undyed pajama check aramid fabric conforming to MIL-C-43774, except that anti-static finish is not required and shall be coated on one side only with a white chloroprene coating. The finished coated fabric shall conform to the physical requirements shown in Table I when tested as specified in 4.4.1 (see 6.5).

TABLE I - Physical Requirements

Characteristic	Requirement Minimum	Maximum
Weight (oz/yd^2)	9.0	13.0
Breaking Strength (lbs)		
Warp	90.0	
Filling	75.0	
Tearing Strength (lbs)		
Warp	6.0	
Filling	5.0	
Hydrostatic Resistance (lbs/in^2)	25.0	
Flame Resistance		
After Flame (sec)		2.0
Char Length (inches)		4.0

3.3.3.1 Leg protector moisture barrier. The cloth for the leg protector moisture barrier shall be as specified in 3.3.3.

3.3.4 Cloth, abrasion resistant. The fabric used for the structural trouser protective knee patches, the leg tabs and for binding the leg bottoms shall be constructed of a 100% para-aramid yarn, 2/1 right hand warp face twill, with a polymer blend coating on both sides of the fabric to finish with 12 ounces per square yard on the face side (grey shade) and 1.8 ounces per square yard on the back side. The finished coated fabric shall conform to the physical requirements specified in table II when tested as specified in 4.4.1 (see 6.6).

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Table II - Physical Requirements

Characteristic	Requirement Minimum	Maximum
Weight (oz/yd^2)	18.0	23.0
Thickness (inches)	0.020	0.030
Breaking Strength (lbs)		
Warp	400.0	-
Filling	325.0	-
Tearing Strength (lbs)		
Warp	45.0	-
Filling	45.0	-
Flame Resistance Vertical		
After Flame (sec)	-	2.0
Char Length (inches)	-	4.0
Abrasion Resistance, 2000 cycles	-	1/

1/ There shall be no visual exposure of the base fabric.

3.3.5 Crotch reinforcement. The para-aramid twill (8.0 oz/yd^2) material used for reinforcing the crotch on the proximity trouser cover, shall be Southern Mills Style # 500KT - 2/1 or equal (see 6.7 and 6.13).

3.3.6 Thread. The thread for assembling the trouser shall be a 60 Tex aramid, natural shade, conforming to A-A-50195. As an alternate, a para-aramid thread conforming to size T-50 of MIL-T-44100, natural in color may be used.

3.3.7 Heat sealing tape. The 7/8 inch wide clear tape used for heat sealing the seams on the moisture barrier material, shall be Electro Seal Corp. Style # ST182 or equal (see 6.8 and 6.13).

3.3.8 Reflective tape. The reflective tape shall be in accordance with 3M style 8487 fluorescent lime/yellow, 3 inch wide or equal (see 6.9 and 6.13).

3.3.9 Hook and loop fastener tape. The hook and loop fastener tape shall conform to type II, class 1, of MIL-F-21840. The tape shall be of 5/8 inch, 1 inch, or 2 inch widths as specified in Tables III and IV. The color shall be either natural, white or black.

3.3.10 Elastic tubing and connectors.

3.3.10.1 Elastic tubing. The seamless tubing used in the leg protector shall conform to types I or V, classes 2 or 3 of ZZ-T-831. Any color of tubing may be used.

3.3.10.2 Tubing connector. The tubing connectors used to join the two ends of the tubing shall be straight nylon hose fittings in accordance with McMaster Carr Supply Co. style # 5380-K12 or equal (see 6.10 and 6.13).

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3.3.11 Suspenders. The heavy duty suspender assembly (webbing, swivel leather fasteners, nickel plated steel slide, rivet & caps, etc.), used on the structural trousers, shall be Body Guard (Division of Lion Apparel) Style #SR100 or equal (see 6.11 and 6.13).

3.3.12 Hardware.

3.3.12.1 Fasteners, snap (standard). The snap fasteners for attaching the snap-in liner and for attaching the proximity trouser cover with the exception of the left fly socket stud combination, shall consist of a button and socket (female half) and a stud and eyelet (male half). The fasteners shall be style 2, 24-line (size 1), nickel plate or bright chrome finish conforming to MS-27980 and the following part numbers:

Button - 1N or 1C	Socket - 6N or 6C
Stud - 7N or 7C	Eyelet - 8N or 8C

3.3.12.2 Fasteners, snap (special). The combination stud and socket fastener used on the left fly of the structural trouser shall be manufactured of nickel plated brass in accordance with Scovill #393 stud, #588 socket, #722 post, and #452 snap or equal (see 6.12 and 6.13).

3.3.12.3 Eyelets and washers. The eyelets and washers for the pocket drain holes shall conform to rolled flange type, M20652/1-BBE114 and M20652/1-BBW101 of MIL-E-20652.

3.3.13 Labels.

3.3.13.1 Label, Structural trouser. Each structural trouser shall have a combination size, instruction and identification label conforming to type VI, class 14 of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

TROUSERS, FIREMEN'S, STRUCTURAL, (TYPE I)
FIBER CONTENT:
Outershell - 60% Para-aramid/40% PBI
Moisture Barrier - Aramid Chloroprene Coated
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE/LENGTH: 38/REGULAR (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

LAUNDERING INSTRUCTIONS
HAND WASH IN LUKE WARM WATER
DO NOT BLEACH
HANG TO DRY
DO NOT MACHINE WASH OR DRY CLEAN

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WARNING

STRUCTURAL TROUSER MUST BE WORN WITH LINER INTACT AND ALL CLOSURES FASTENED WHEN IN USE. DO NOT KEEP IN DIRECT CONTACT WITH FLAME. DO NOT STORE THIS GARMENT IN DIRECT SUNLIGHT. KEEP GARMENT CLEAN AS SOILING WILL REDUCE PROTECTIVE QUALITIES. FAILURE TO COMPLY WITH THESE INSTRUCTIONS MAY RESULT IN SERIOUS INJURY OR DEATH.

DO NOT REMOVE THIS LABEL

3.3.13.2 Warning label, structural trouser. Each structural trouser shall have a warning label conforming to type IV, class 9 of DDD-L-20 applied to the inside back moisture barrier. The label marking shall be applied to the moisture barrier by a red marking medium. The label shall be printed with block sansserif type with letters 3/4 to 1 inch in height. The label shall contain the following inscription:

WARNING
MUST BE WORN WITH
LINER IN PLACE

3.3.13.2.1 NFPA approval label, structural trouser. Each structural trouser shall have a NFPA approved label conforming to type VI, class 3, of DDD-L-20. The label shall contain the following inscription:

THIS STRUCTURAL PROTECTIVE GARMENT, MEETS THE REQUIREMENTS OF NFPA 1971 STANDARD ON PROTECTIVE CLOTHING FOR STRUCTURAL FIREFIGHTING, 1991 EDITION.

WARNING

FOR STRUCTURAL FIREFIGHTING OPERATIONS, BOTH THE STRUCTURAL COAT AND TROUSER MUST BE WORN OVER THEIR APPLICABLE AND DETACHABLE LINERS FOR LIMB/TORSO PROTECTION. OUTER SHELL, MOISTURE BARRIER, AND THERMAL BARRIER MEETING ALL REQUIREMENTS OF NFPA 1971 MUST BE UTILIZED AND ALL GARMENT CLOSURES MUST BE FASTENED WHEN IN USE. DO NOT KEEP THIS GARMENT IN DIRECT CONTACT WITH FLAMES. THIS GARMENT ALONE MAY NOT PROVIDE PROTECTION FOR FIRE ENTRY APPLICATIONS OR FOR PROTECTION FROM CHEMICAL, BIOLOGICAL, OR RADIOLOGICAL AGENTS. OTHER PROTECTIVE EQUIPMENT (HEAD PROTECTION, SCBA, GLOVES, FOOTWEAR, PASS) IS REQUIRED FOR PROTECTION. KEEP THIS GARMENT CLEAN AS SOILING WILL REDUCE PROTECTIVE QUALITIES.

DO NOT USE CHLORINE BLEACH

CHLORINE BLEACH WILL SIGNIFICANTLY COMPROMISE THE PROTECTION AFFORDED BY THE TEXTILE MATERIALS UTILIZED IN THE CONSTRUCTION OF THE GARMENT. USERS MUST CLEAN, MAINTAIN, AND ALTER ONLY IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. DO NOT STORE IN DIRECT SUNLIGHT. NO PROTECTIVE CLOTHING CAN PROVIDE COMPLETE PROTECTION FROM ALL CONDITIONS. USE EXTREME CARE FOR ALL EMERGENCY OPERATIONS. FAILURE TO COMPLY WITH THESE WARNINGS MAY RESULT IN SERIOUS INJURY OR DEATH.

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DO NOT REMOVE THIS LABEL

THIS STRUCTURAL TROUSER WHEN WORN WITH A LINER COMPLYING WITH MIL-T-24942 AND UNDER THE REFLECTIVE TROUSER COVER COMPLYING WITH MIL-T-24940, MEETS NFPA 1976 STANDARD ON PROTECTIVE CLOTHING FOR PROXIMITY FIREFIGHTING - 1992 EDITION.

Classified by Underwriters Laboratories Inc. in accordance with the National Fire Protection Association Standard for Structural Fire Fighting NFPA 1971-1991.

3.3.13.3 Label, proximity trouser cover. Each proximity trouser cover shall have a combination size, instruction and identification label conforming to type VI, class 14 of DDD-L-20 except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

TROUSER COVER, FIREMEN'S, ALUMINIZED PROXIMITY, (TYPE II)
FIBER CONTENT: - 100% Para-aramid
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE/LENGTH: 38/REGULAR (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER (If other than contractor)

INSTRUCTIONS

1. CAUTION: MUST BE USED ONLY AS A COVER WITH STRUCTURAL TROUSER.
2. Hang to dry and store properly to prevent damage to aluminized surfaces.
3. Clean by gently rubbing surface with cloth using mild soap, PERCHLOROETHYLENE or ISOPROPANOL. When cleaning with perchloroethylene or isopropanol, ensure that you have adequate ventilation and that you use proper gloves and eye protection.
4. To maintain reflective properties and to extend service life, use proximity trouser cover for intended purpose only.
5. Do not use aluminized paint to repair worn out surface as it will not provide protection.

NOTE: Aluminized coating is damaged by contact with fire extinguishing agent Potassium Bicarbonate Powder (PKP). Before storing garment with PKP residue present, the following is recommended.

1. Dry powder should be removed with dry brush or blower.
2. Wet powder should be removed by thoroughly flushing with water and wiping the surface down with a clean soft cloth. Damp or wet garments should be hung to dry.

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3.3.13.4 NFPA approval label, proximity trouser cover. Each cover shall have a NFPA approval label conforming to type VI, class 3, of DDD-L-20. The label shall contain the following inscription:

THIS PROXIMITY PROTECTIVE GARMENT MEETS THE REQUIREMENTS OF NFPA 1976 STANDARD ON PROTECTIVE CLOTHING FOR PROXIMITY FIRE FIGHTING, 1992 EDITION.

WARNING

FOR PROXIMITY FIREFIGHTING OPERATIONS, THIS REFLECTIVE TROUSER COVER MUST BE WORN OVER THE STRUCTURAL TROUSER COMPLYING WITH MIL-T-24940 AND LINER COMPLYING WITH MIL-L-24942. BOTH THE STRUCTURAL COAT AND TROUSER MUST BE WORN OVER THEIR APPLICABLE AND DETACHABLE LINERS FOR LIMB/TORSO PROTECTION. OUTER SHELL, MOISTURE BARRIER, AND THERMAL BARRIER MEETING ALL REQUIREMENTS OF NFPA 1976 MUST BE UTILIZED AND ALL GARMENT CLOSURES MUST BE FASTENED WHEN IN USE. DO NOT KEEP THIS GARMENT IN DIRECT CONTACT WITH FLAMES. THIS GARMENT ALONE MAY NOT PROVIDE PROTECTION FOR FIRE ENTRY APPLICATIONS OR FOR PROTECTION FROM CHEMICAL, BIOLOGICAL, OR RADIOLOGICAL AGENTS. OTHER PROTECTIVE EQUIPMENT (HEAD PROTECTION, SCBA, GLOVES, FOOTWEAR, PASS) IS REQUIRED FOR PROTECTION. KEEP THIS GARMENT CLEAN AS SOILING WILL REDUCE PROTECTIVE QUALITIES.

DO NOT USE CHLORINE BLEACH

CHLORINE BLEACH WILL SIGNIFICANTLY COMPROMISE THE PROTECTION AFFORDED BY THE TEXTILE MATERIALS UTILIZED IN THE CONSTRUCTION OF THE GARMENT. USERS MUST CLEAN, MAINTAIN, AND ALTER ONLY IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. DO NOT STORE IN DIRECT SUNLIGHT. NO PROTECTIVE CLOTHING CAN PROVIDE COMPLETE PROTECTION FROM ALL CONDITIONS. USE EXTREME CARE FOR ALL EMERGENCY OPERATIONS. FAILURE TO COMPLY WITH THESE WARNINGS MAY RESULT IN SERIOUS INJURY OR DEATH.

DO NOT REMOVE THIS LABEL

Classified by Underwriters Laboratories Inc. in accordance with the National Fire Protection Association Standard for Proximity Fire Fighting NFPA 1976-1992.

3.3.13.5 Size label, proximity trouser cover knee patch. Each knee patch shall have a size label conforming to type IV, class 2 of DDD-L-20. The label marking shall be applied to the back side of the knee patch by a red marking medium. The label shall be printed with block sansserif type with letters 3/4 to 1 inch in height. The label shall contain the following inscription:

SIZE/LENGTH: 38 R (EXAMPLE)

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3.3.14 Repair of aluminum film. The aluminum film used for patches for repairing uncoated or cracked areas on the finished proximity trouser cover shall be the same as specified for coating the base cloth. The area to be repaired shall not exceed 1 inch in the longest direction. The patch shall be completely bonded onto the aluminum film of the proximity trouser cover and shall be so placed that it extends not less than 1/2 inch beyond all edges of the uncoated or cracked area. No outer shell shall contain more than 3 patches. The unsupported aluminum film and self-curing adhesive used for repairing shall be obtained from the basic material supplier (see 3.3.2). The size of the patch shall not exceed 2 inches in any direction.

3.3.15 Instruction manual. An instruction manual shall be furnished by the contractor with each structural trouser and proximity trouser cover. For the structural trouser, the manual shall be placed into the cargo pocket (see 5.1.1.1). For the proximity trouser cover, the manual shall be inserted into the polyethylene bag during packaging, and positioned so that it is visible (see 5.1.1.2). The printing, collation, folding and stitching of the manual shall be in accordance with the standard sample (see 6.3). The cover shall be antique white book paper conforming to Joint Committee Printing-A-100, 140 pound basis weight 25 X 38 inches per 1000 sheets. The text paper shall be white offset book paper conforming to JCP-A-60, 120 pound basis weight 25 X 38 inches per 1000 sheets. A sample of the manual, plus four glossy pictures used in printing of the instruction manual, shall be obtained from the contracting officer.

3.4 Design. The trouser shall be of a two piece design, consisting of an inner structural trouser and a detachable outer proximity trouser cover. The structural trouser shall be fabricated from a para-aramid/PBI fabric outer shell and lined with a chloroprene coated aramid fabric for the moisture barrier. The detachable outer proximity trouser cover shall be fabricated from a highly reflective aluminized para-aramid knit fabric. The structural trouser shall have a fastener tape and snap fastener front closure, detachable heavy duty suspenders secured to the waistband at front and back, and a strip of reflective tape around each lower leg. The knees are protected by patches made of abrasion resistant cloth. The right leg has a cargo pocket and flap. The outer proximity trouser cover is designed to fit over the structural trouser. Attachment is made by snap fasteners at the waist and the fly is closed by means of fastener tape. The inside bottom of each leg is fitted with an elasticized protector. There are 4 additional sets of knee patches included with each proximity trouser cover (see Figures 1 and 2).

3.4.1 Figures. Figures are furnished for informational purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification. The patterns allow a 1/2 inch seam allowance for major joining seams, except as otherwise indicated in Tables III and IV.

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3.5.1 Pattern parts. The component parts of the trousers shall be cut from the materials specified and in accordance with the number of parts required for the manufacturing process as follows:

MATERIAL	NOMENCLATURE OF PATTERN PARTS	CUT PARTS
	STRUCTURAL TROUSERS (TYPE I)	
Base fabric	Front	2
	Back	2
	Waistband facing	4
	Left fly	1
	Right fly	2
	Cargo pocket	1
	Cargo pocket flap	1
Moisture barrier	Front	2
	Back	2
	Right fly interlining	2
	Cargo pocket reinforcement	1
	Cargo pocket flap reinforcement	1
Abrasion resistant cloth	Knee patch	2
	PROXIMITY TROUSER COVER (TYPE II)	
Aluminized	Front	2
	Back	2
	Knee patch holder	2
	Knee patch	10
	Waistband facing	2
	Left fly extension	1
Para-aramid twill	Front crotch piece	2
	Back crotch piece	2
Moisture barrier	Leg protector	2
Templates	Fastener tape location marker	1

3.6 Construction.

3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types specified in Tables III and IV shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation any one may be used. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the garment. Topstitching, edgestitching and staystitching shall be 1/16 to 3/16 inch from the edge, hook and loop fastener tape and reflective tape stitching shall be 1/16 to 3/16 inch from the tape edge. All double needle stitching shall be 1/4 inch gauge, with the row of stitching next to the folded edge not more than 1/16 inch from the edge. The bight of the overedge stitching shall be 3/16 to 5/16 inch. A knife shall be used on the overedge machine with the guide set so that only the ravelled edges are trimmed. In order to minimize the size of the needle hole in the aluminized fabric, the needle used in the seaming operations shall be the smallest size which will accommodate the specified size thread.

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3.6.2 Thread breaks and end of seams. Ends of all seams and stitching when not caught in other seams or stitching shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch. The ends of label stitching shall be overlapped not less than 3 stitches. Thread breaks (all stitch type) shall be repaired by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired by using 301 stitch type, except for seat seam which shall be resewn. When ends of overedge stitching are not caught in other seams or stitching, the ends shall be tacked. When stitch type 401 is permitted as an option, it shall not be used unless the ends are caught in other seams of stitching.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Tables III and IV.

3.7 Manufacturing operations requirements. The trousers shall be manufactured in accordance with the operation requirements specified in Tables III and IV. The supplier is not required to follow the exact sequence of operations listed provided that the finished trouser is identical to those produced by following the sequence of operations as listed in Tables III and IV.

3.7.1 Size marking. The component parts of the trousers shall be bundled, marked or ticketed to assure proper size assembly. The use of metal fastening devices or sewn-on tickets for marking is prohibited. Adhesive tickets, when used, shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machines, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the trousers and wherever possible, is covered by the seam allowance.

3.7.2 Abbreviations in table of operations. The abbreviations used in tables III an IV are as follows:

Stch - Stitch
In - Inch
Ndl - Needle
Bob - Bobbin
Lpr - Looper
Brtk - Bartack

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL BOB, LPR
1.	Cutting.				
	a. Cut the structural trousers in strict accordance with patterns which show directional lines, marks and notches for proper assembly. Directional lines shall not appear on the fabrics. Drill holes are not permitted except to aid in the attachment of hardware.				
	b. The component parts of the base fabric may be cut from ends and when so cut shall approximate the color of the major cut parts.				
	c. The component parts of the moisture barrier may be cut from ends.				
	d. Cut the leg tabs from the abrasion resistant cloth, to finish 1 inch wide by 4 (±1/4) inches in length (see 3.3.4).				
	e. The face side of the materials to be assembled shall be considered as follows: Outershell - as marked Moisture barrier - uncoated side Abrasion resistant cloth - grey coated side				
	NOTE: During the manufacturing of this garment, the water repellent coated side of the moisture barrier shall face away from the wearer in the finished structural trouser.				
2.	Replacement of defective components.				
	During the spreading, cutting and manufacturing operations, components having material defects or damages that are classified as damages in MIL-STD-1666 and section 4 shall be removed and replaced with nondefective and properly matched components.				

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
3.	Seam direction.					
	The direction of the joining seams shall be lapped as follows: 1. Front over back at side seams and inseams. 2. Left over right at seat seam and crotch seam.					
4.	Make waistband facing.					
	a. Position waistband facings, face to face, at center back notch and stitch 1/2 inch from the raw edge. Spread seam allowance open and flat. NOTE: For both inner and outer plies of waistband facings.	301	SSa-1	6-8	B	B
	b. Position inner and outer plies of waistband facings, face to face, and stitch 1/4 inch from raw edge of the ends and bottom edge of waistband facing.	301	SSe-2(a)	6-8	B	B
	c. Turn waistband facing to finished position, work out corners and edges and topstitch turned edges.	301	SSe-2(b)	6-8	B	B
	d. For ease in manufacturing, the raw edges may be staystitched together, 1/4 inch from the raw edge.	301	SSa-1	6-8	B	B
	e. Center and attach nine snap fastener sockets and buttons to the lower section of the facing as indicated by the pattern marks (with sockets facing away from wearer as worn).					
5.	Make right fly.					
	a. Overedge the straight edge of the underply of the right fly .	504	EFd-1	6-8	B	B
	b. Position a strip of 1 inch wide hook fastener tape to outerply of right fly as indicated by marks on pattern. Stitch along all edges of the tape.	301 or 401	LSbj-1	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
5.	Make right fly. (cont'd)					
	c. Position the two right fly pieces face to face and superimpose both right fly interlinings and stitch the curved edge 1/4 inch from the raw edge through all plies.	301	SSa-1	6-8	B	B
6.	Prepare left fly piece.					
	a. Overedge the back curved edge of left fly.	504	EFd-1	6-8	B	B
	b. Position a strip of 1 inch loop fastener tape to the face of the left fly piece as indicated by marks on patterns. Stitch along all edges of the tape.	301	LSbj-1	6-8	B	B
7.	Make cargo pocket.					
	a. Position cargo pocket reinforcement to back side of cargo pocket. The two plies may be staystitched around outer edges.	301	SSa-1	6-8	B	B
	b. Position three metal eyelets on cargo pocket in accordance with marks on patterns and attach with the center of each eyelet 1 (±1/8) inch from the finished bottom edge of the pocket.					
	c. Turn under the top raw edge of the pocket 1/2 inch and fold top as indicated by marks on patterns forming a 2-3/4 inch hem. Stitch 1/8 inch and again 2-5/8 inch from top folded edge.	301	EFt-2	6-8	B	B
	d. Position two 1-1/2 by 3 inch pieces of hook fastener tape on the cargo pocket as indicated by marks on patterns. Stitch along all edges.	301	LSbj-1	6-8	B	B
	e. To make pleats fold front edge of cargo pocket on pleat fold line closest to the raw edge, face to face, and stitch 1/8 inch from folded edge.	301	OSf-1	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
7.	Make cargo pocket. (Cont'd)					
	f. Fold edge of cargo pocket on second pleat fold line, back to back, and stitch 1/8 inch from folded edge.	301	OSf-1	6-8	B	B
8.	Make cargo pocket flap.					
	a. Position two 1-1/2 by 3 inch pieces of loop tape on the underside of the cargo pocket flap as indicated by marks on patterns. Stitch along all edges.	301	LSbj-1	6-8	B	B
	b. Position the cargo pocket reinforcement on the inside of the top layer of the cargo pocket flap and staystitch together along both side edges.	301	SSa-1	6-8	B	B
	c. Fold cargo pocket flap in half, face to face, as indicated by marks on patterns and stitch 3/8 inch from the raw side edges.	301	SSe-2(a)	6-8	B	B
	d. Trim points, turn cargo pocket flap to finished position and completely force out corners. Topstitch the turned in sides and bottom folded edge.	301	SSe-2(b)	6-8	B	B
	e. Overedge raw edges of cargo pocket flap.	504	EFd-1	6-8	B	B
9.	Attach reflective tape and knee patches to legs.					
	a. Position reflective tape to lower legs in accordance with marks on patterns. The tape shall be parallel to the bottom edges of the legs. Topstitch each side of the reflective tape to the legs.	301	LSbj-1	6-8	B	B
	or					
	b. As an alternate, after stitching outseam of trouser (oper.11a) position reflective tape to front and back trouser assembly according to marks on patterns and stitch as in (oper.9a).	301	LSbj-1	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
9.	Attach reflective tape and knee patches to legs. (cont'd)					
	c. Position abrasion resistant cloth knee patches to fronts of legs as indicated by marks on patterns. Double stitch 1/4 inch gauge with outer row of stitching 1/8 to 1/4 inch from the raw edge of the abrasion resistant cloth patch.	301 or 401	LSbj-2	6-8	B	B
10.	Prepare moisture barrier assembly and attach labels.					
	a. Stitch 1/2 inch from raw edge, the seat seam, crotch seam, outseam and inseam, face to face.	301 or 401	LSq-2(a)	6-8	B	B
	b. Turn one ply back sharply over the first row of stitching and stitch 1/8 (±1/16) inch from the folded edge.	301	LSq-2(b)	6-8	B	B
	or					
	c. As an alternate, operations 10.a and 10.b may be performed in one operation with a double lapped doubled stitched seam, 1/4 inch gauge.	301 or 401	LSc-2	6-8	B	B
	d. Position and topstitch around all sides a combination label (see 3.3.13.1) to moisture barrier on inside back of trousers, approximately 1 inch to the right of the center seam and 1 to 2 inches below the bottom edge of waistband facing.	301	LSbj-1	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
10.	Prepare moisture barrier assembly and attach labels (cont'd).					
	e. Apply the warning label, (see 3.3.13.2) to moisture barrier on inside back of trousers, approximately 1 inch to the left of the center seam and 1 to 2 inches below the bottom edge of waistband facing.					
	f. Position and topstitch around all sides the NFPA approval label (see 3.3.13.2.1) to moisture barrier on inside back of trousers, approximately 1 inch to the right of the center seam and 1 inch below the bottom edge of the combination label.	301	LSbj-1	6-8	B	B
	g. Each of the moisture barrier seams on coated side of fabric, shall be heat sealed with the 7/8 inch wide tape specified in 3.3.7.					
11.	Prepare outershell and attach pockets and flaps.					
	a. Join outseams with a double lapped and double stitched seam, catching and matching the ends of the reflective tape.	301 or 401	LSc-2	8-10	B	B
	b. Turn raw pleated front edge of cargo pocket under 1/2 inch. Lay cargo pocket flat with pleats closed. Turn bottom and back raw edge under 1/2 inch and position on right leg as indicated by marks on pattern. Topstitch both sides and bottom edges. Place a second row of topstitching along bottom and back edge 1/4 inch inside of first topstitching.	301 or 401	LSd-1 and LSd-2	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
11.	Prepare outershell and attach pockets and flaps. (cont'd)					
	c. The finished cargo pocket for all sizes shall measure 9-3/4 (±3/8) inches long at center. The width of the pocket shall be 8 (±3/8) inches wide when measured across top edge of pocket from turned edge of pleat to back turned edge.					
	NOTE: Care shall be exercised so as not to catch the outer pleated edge of pocket in the stitching of the pocket front edge.					
	d. Turn under 1/4 inch, the overedged top of the cargo pocket flap and double stitch 1/4 to 5/16 inch gauge to right leg as indicated by marks on patterns. The outer row of topstitching shall be 1/8 (±1/16) inch below top folded edge of the pocket flap.	301 or 401	LSba-2	6-8	B	B
	or					
	e. As an alternate, position the cargo pocket flap face down on leg, with raw edges facing towards the bottom of the leg and 3/8 inch below the mark on the patterns. Stitch 1/4 inch from the raw edge. Turn pocket flap to finished position and double stitch 1/4 to 5/16 inch gauge with outer row of stitching 1/8 (±1/16) inch below top folded edge of pocket flap.	301 or 401	LSbl-3	6-8	B	B
	f. The finished cargo pocket flaps for all sizes shall measure 8 (±1/4) inches wide and 3-3/4 (±1/4) inches long.					

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
11.	Prepare outershell and attach pockets and flaps. (Cont'd)					
	g. Stitch a 5/8 inch vertical bartack at the top corners of the cargo pocket, a 5/8 inch horizontal bartack at the bottom front corner of the cargo pocket and a 5/8 inch horizontal bartack at each top corner of the cargo pocket flap. The bartacks shall be superimposed on the previous inner line of stitching.	Brtk		42 per Brtk	B	B
	h. Join the crotch seam, seat seam, and inseam with a double lapped and double stitched seam, catching and matching the ends of the reflective tape.	301 or 401	LSc-2	8-10	B	B
12.	Join shell and moisture barrier.					
	Insert moisture barrier into outershell with coated side of moisture barrier away from the body as worn. Staystitch around top and fly edge 1/4 inch from the raw edge.	301	SSa-1	6-8	B	B
13.	Attach waistband facing and fly pieces.					
	a. Position raw edge of left fly, face to face, to left front trouser. Stitch front edge of fly 1/4 inch from the raw edge, through all plies. Clip trouser seam allowance at bottom of fly.	301	SSa-1	6-8	B	B
	b. Turn left fly to finished position and stitch back edge to left front 3/8 inch from overedged back edge, forming the "J" stitching.	301	Similar to LSag-3	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
13.	Attach waistband and fly pieces. (Cont'd)					
	c. Position the raw edge of the outerply of the right fly and interlining assembly, face to face, to the right front trouser fly opening. Stitch 1/4 inch from raw edge through all plies.	301	SSe-2(a)	6-8	B	B
	NOTE: Clip seam allowance of fly opening to stitching.					
	d. Stitch across raw top edges of right fly piece, face to face, 3/8 inch from raw edge.	301	SSe-2(a)	6-8	B	B
	e. Trim corner and turn right fly assembly, forcing out corner, to finished position with trouser seam allowances flat and turned inside the right fly assembly. Topstitch the turned seam allowances of the outerply of the right fly.	301	SSe-2(b)	6-8	B	B
	f. Topstitch the back edge of right fly piece through all plies, making sure seam allowances are flat.	301	LSs-2(b)	6-8	B	B
	g. Position waistband facing to top edge of trousers, face to face, so left edge of waistband facing and fly are even. Stitch 3/8 inch from the raw edge the entire length of waistband facing.	301	SSe-2(a)	6-8	B	B
	h. Turn waistband facing to finished position and topstitch across top edge of trousers and front edges of left and right fly pieces. Make certain that top stitching of left fly continues onto crotch closing seam.	301	SSe-2(b)	6-8	B	B
	i. Stitch down waistband facing through all plies, 1-3/4 (±1/8) inch from top finished edge of trousers. Stitching shall extend from left front edge to beginning of right fly piece.	301	Similar to LSag-3	6-8	B	B

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
13.	Attach waistband and fly pieces. (Cont'd)					
	j. With front edges flat and smooth place a 1/2 to 5/8 inch horizontal bartack at base of fly securing both right and left fly pieces.	Brtk		26-30 per Brtk	B	B
14.	Finish off top of trousers.					
	a. Attach seven snap fasteners on outside of waistband as follows:					
	1. Attach special snap fastener stud and socket combination to left fly, positioned 3/4 inch from top edge of waistband and from outside edge of left fly to center of stud. The stud shall face away from the wearer as worn.					
	2. Attach snap fastener stud and eyelet combination to right fly, positioned 3/4 inch from top edge of waistband and from seam edge of fly to center of stud. The stud shall face away from the wearer as worn, and shall align with the socket on left fly when the fly is closed.					
	3. Attach snap fastener stud and eyelet combinations to waistband on front, positioned 3/4 inch from top edge of waistband. The snap fastener combinations shall be positioned 5-1/2 inches from the outside edge of the left fly and 7 inches from the outside edge of the right fly. The studs shall face away from the wearer as worn.					
	4. Attach snap fastener stud and eyelet combinations to back of waistband, positioned 3/4 inch from top edge of waistband and 3-1/2 inches from side seam to center of stud. The studs shall face away from the wearer as worn.					

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL BOB, LPR
14.	Finish off top of trousers. (cont'd)				
	5. Attach snap fastener stud and eyelet combination to center back seam of waistband, positioned 3/4 inch from top edge of waistband to center of stud. The stud shall face away from the wearer as worn.				
	Note: Each of the eyelets on snap fastener assembly should be reinforced with a piece of material cut from the abrasion resistant cloth (see 3.3.4)				
	b. Attach eight (rivet and cap) combinations for suspender assembly on outside of waistband as follows:				
	1. Two assembled rivets and caps shall be positioned on the left and right fronts of the trouser to accommodate the suspenders and the swivel leather fastener. The distance between the front rivet/cap assembly shall be 3 inches, with the snap fastener stud centered between the two leather fasteners and 3/4 inch from top edge of waistband to center of stud.				
	2. Four assembled rivets and caps shall be positioned on back of trouser to accommodate the remaining suspender straps and leather fasteners. The distance between the back rivet/cap assemblies shall be 3 inches equally spaced from center back seam and 3/4 inch from top edge of waist band to center of rivet.				
	3. Attach the suspender assembly to the trousers.				

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TABLE III - CONSTRUCTION OF STRUCTURAL TROUSERS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
15.	Finish bottom edge of trousers.					
	a. Center and attach a snap fastener socket and eyelet combination to the end of the abrasion resistant cloth leg tab. The center of the socket shall be 1/2 inch from the end of the abrasion resistant cloth leg tab.					
	b. Position to each inside leg seam, a leg tab with the socket end towards top of trouser and socket facing towards moisture barrier. The leg tab will extend to the bottom of the hem and be included in the hem stitching. Bartack or tack to trouser leg through all plies full width of tab and 1/4 (±1/8) inch from finished top edge of leg hem.	301 or Brtk	LSbj-1	6-8 26-30 per Brtk	B	B
	c. Turn under entire bottom edge of outershell 3/8 to 1/2 inch and form a 1-1/2 (±1/8) inch wide hem. Stitch top and bottom edges of hem 1/8 to 3/16 inch from each turned edge. Seams of moisture barrier and proximity trouser cover shall match up. NOTE: The bottom raw edge of the moisture barrier shall extend to the fold of the outershell hem and should be caught in stitching of finished hem.	301	Similar to EFq-2	6-8	B	B
16.	Clean trousers.					
	a. Trim all thread ends.					
	b. Remove all loose threads.					
	c. Remove all spots and stains.					

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TABLE IV - CONSTRUCTION OF PROXIMITY TROUSER COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL BOB, LPR
1.	Cutting.				
	a. Cut the proximity trouser cover in strict accordance with patterns which show directional lines, marks and notches for proper assembly. Directional lines shall not appear on the fabric. Drill holes are not permitted.				
	b. The component parts of the aluminized fabric may be cut from ends and when so cut shall finish with the aluminized side face up.				
	c. Cut the strips for binding the bottom edges of leg openings from the abrasion resistant cloth, to finish 2 inches in width (see 3.3.4).				
	d. Cut 15 inch lengths of elastic rubber tubing and connect ends together with tubing connectors.				
	e. The face side of the proximity trouser cover material is the aluminized side.				
	f. The face side of the leg protector material is the uncoated side.				
2.	Replacement of defective components.				
	During the spreading, cutting and manufacturing operations, components having material defects or damages that are classified as damages in MIL-STD-1666 shall be removed and replaced with nondefective and properly matched components.				

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TABLE IV - CONSTRUCTION OF PROXIMITY TROUSER COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
3.	Seam direction.					
	The lapping of the joining seams shall be as follows:					
	1. Fronts over back at side seams and inseams.					
	2. Left over right at seat seam and crotch seam.					
4.	Assemble leg protector.					
	a. Fold leg protector, end to end, face to face and stitch 3/8 inch from raw edges.	301	LSq-2(a)	6-8	B	B
	b. Fold top edge of leg protector as indicated by the pattern marks, insert elastic leg tubing and stitch 3/16 inch from the raw edge enclosing the elastic in the leg protector. Care shall be taken not to catch the elastic in the stitching.	301	EFa-1	6-8	B	B
5.	Make knee patch and knee patch holder.					
	a. Fold top edge of knee patch face to face 1/2 inch, and position a strip of 5/8 inch loop fastener tape to finish on face side across the width of the knee patch, simultaneously folding each side of knee patch face to face 1/2 inch. The edge of the fastener tape shall be even with the folded edges of the knee patch. Stitch around edge of loop fastener tape.	301	LSbj-1	6-8	B	B

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TABLE IV - CONSTRUCTION OF PROXIMITY TROUSER COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
5.	Make knee patch and knee patch holder (cont'd).					
	b. Form pleats in both sides of knee patch as indicated by notches. The fold of the pleat must face toward the bottom of the knee patch. Depth of pleat shall be 3/8 inch.					
	c. Fold over side edges of knee patch, with pleats face to face 1/2 inch, and position a strip of 5/8 inch loop fastener tape from the top to the bottom edge of knee patch, simultaneously folding bottom of knee patch 1/2 inch back to back on each side. The edge of each fastener tape shall be even with the folded edge of the knee patch. Stitch around edge of loop fastener tape.	301	LSbj-1	6-8	B	B
	d. Position a strip of 5/8 inch loop fastener tape to back side of knee patch across bottom. The edge of the fastener tape shall be even with the folded edge of the knee patch. Stitch around edge of the loop fastener tape.	301	LSbj-1	6-8	B	B
	e. Clip inside corners and fold under the inside edges of knee patch holder 1/2 inch. Position a strip of 5/8 inch hook fastener tape to back side of knee patch holder. Bottom edges of knee patch holder shall be turned under 1/2 inch. The edge of the fastener tape shall be even with the folded inner edges of knee patch holder. Stitch around edges of hook fastener tape.	301	LSbj-1	6-8	B	B
	f. Cut and attach a strip of 5/8 inch hook fastener tape to proximity trouser cover front in accordance with marks on pattern. Stitch around edges of hook fastener tape.	301	LSbj-1	6-8	B	B

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TABLE IV - CONSTRUCTION OF PROXIMITY TROUSER COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
6.	Attach knee patch holder to knee.					
	a. Position knee patch holder to proximity trouser cover front in accordance with notches on pattern, with angled edge aligned evenly with inseam. Turn under 1/2 inch top edge of knee patch holder, and stitch assembly to proximity trouser cover front 1/8 inch from folded edge. NOTE: The sides of the knee patch holder may be staystitched to the proximity trouser cover front 1/8 inch from the raw edge.	301	LSb-1	6-8	B	B
	b. Insert the knee patch within the knee patch holder.					
7.	Assemble proximity trouser cover.					
	a. Staystitch front crotch pieces to front trousers along crotch seam and inseam, matching notches.	301	SSa-1	6-8	B	B
	b. Topstitch outer curve of front crotch pieces to front trousers.	301	LSbj-1	6-8	B	B
	c. Staystitch back crotch pieces to back trousers along seat seam and inseam.	301	SSa-1	6-8	B	B
	d. Topstitch outer curve of back crotch pieces to back trousers.	301	LSbj-1	6-8	B	B
	e. Join the outseam, inseam, crotch seam and seat seam using a double lapped and double stitched seam.	301 or 401	LSc-2	8-10	B	B
	or					
	f. As an alternate, join the outseam, inseam, crotch seam and seat seam. Turn seam allowance to one side and topstitch 1/8 to 3/16 inch from the turned edge.	301	LSq-2	6-8	B	B
	g. Position waistband facing, face to face, at center back notch and stitch 1/2 inch from the raw edge.	301	SSa-1	6-8	B	B

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
7.	Assemble proximity trouser cover (cont'd).					
	h. Position waistband facing on trouser, face to face, matching the center back seams and stitch 3/8 inch from raw edges around top.	301 or 401	SSe-2(a)	6-8	B	B
	i. Turn waistband facing to finished position and topstitch 1/4 inch from the top folded edge of the trousers, through all plies. Seam of waistband facing shall not show from outside of garment.	301	SSe-2(b)	6-8	B	B
	j. Stitch waistband facing through all plies 1-3/4 inch from the finished top edge of waistband facing.	301	Similar to LSag-2	6-8	B	B
	k. Bind the right and left fly openings in one continuous operation with a 1-1/2 inch wide strip of aluminized fabric. The finished side of the binding shall finish on the outside of the trouser. Binding shall finish even with the top edge of the trousers and should be 1/2 inch wide.	301	BSb-1	6-8	B	B
8.	Assemble and attach left fly extension.					
	a. Fold left fly extension piece face to face in accordance with the marks on the pattern. Stitch 1/2 (±1/16) inch from the bottom and top raw edges.	301	SSe-2(a)	6-8	B	B
	b. Turn left fly extension piece to the finished position and work out the corners. Topstitch 3/16 (±1/16) inch from the top, side and bottom finished edges.	301	SSe-2(b)	6-8	B	B

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
8.	Assemble and attach left fly extension (cont'd).					
	c. Position a strip of 1 inch wide loop fastener tape on the left fly extension along inside folded edge to finish 1/8 inch from top, bottom and folded side of left fly extension. Stitch along all edges of the tape.	301	LSbj-1	6-8	B	B
	d. Position the raw edge of the left fly extension piece 3/8 inch past the stitching of the bound edge of the left fly opening, with the wide end of the left fly extension piece even with the finished top edge of the trousers. Stitch 3/8 to 7/16 inch from the finished edge of the left fly opening through all plies.	301	LSa-1	6-8	B	B
	e. Position a strip of hook fastener tape on the outside of the right front trouser in accordance with template marker. Marker defines outer edge of tape and shall be positioned against fold of binding edge and top edge of finished proximity trouser cover as indicated. Stitch along all edges of tape. The left fly shall overlap the right fly with a smooth appearance when closed.	301	LSbj-1	6-8	B	B
	f. Stitch a 5/8 inch horizontal bartack, centered over binding seam, catching both plies of binding and left fly extension. Bartack should match topstitching of left fly extension.	Brtk		26-30 per Brtk	B	B
9.	Attach snap fasteners.					
	a. Attach six snap fastener buttons and sockets to the waistband as follows. The sockets on the inside of the cover shall face the wearer:					
	1. Attach button and socket to waistband on left front, positioned 7-1/2 inches from the finished edge of extension piece and 3/4 inch from top edge of waistband to center of socket.					

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
9.	Attach snap fasteners (cont'd).					
	2. Attach button and socket to waistband on left front, positioned 5/8 inch from the bound edge of left fly opening and 3/4 inch from top edge of waistband to center of socket.					
	3. Attach button and socket to waistband on right front, positioned 5-1/2 inches from the bound edge of right fly opening and 3/4 inch from top edge of waistband to center of socket.					
	4. Attach button and socket to waistband on right and left backs, positioned 3/4 inch from top edge of waistband to center of socket and 3-1/2 inches from side seam to center of socket.					
	5. Attach button and socket to center back seam of waistband, positioned 3/4 inch from top edge of waistband to center of socket.					
	Note: Each of the sockets on snap fastener assembly, should be reinforced with a piece of material cut from the abrasion resistant cloth (see 3.3.4).					
10.	Insert leg protector assembly.					
	a. Insert leg protector assembly into the leg, back to back matching joining seam with leg inseam and stitch 1/4 inch from the raw bottom edges.	301	SSa-1	6-8	B	B
	b. Doublestitch through all plies the leg protector assembly to trouser leg, according to the pattern marks. Stitching shall start and stop at the inseam.	301	SSv-2	6-8	B	B

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TABLE IV - CONSTRUCTION OF PROXIMITY TROUSER COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	Thread NDL	Thread BOB, LPR
11.	Bind leg bottoms.					
	Fold abrasion resistant cloth strip in half lengthwise. Stitch each leg opening with a double row of stitching 1/8 to 1/4 inch from raw edges of abrasion resistant cloth strip, through all plies of material, overlapping ends of strip approximately 1 to 1-1/2 inches near the inseam. The gauge of stitching shall be 1/4 inch. The bottoms of the trouser legs shall extend within and to the folded edge of the abrasion resistant cloth strips.	301	BSa-2	6-8	B	B
	NOTE: Folder can be used to perform this operation.					
12.	Labels.					
	a. Position and topstitch the combination label, (see 3.3.13.3) on inside of trouser 1 inch to the right of the center back seam as worn and 1 to 2 inches below bottom raw edge of waist facing.	301	LSbj-1	6-8	B	B
	b. Position and topstitch the NFPA approval label (see 3.3.13.4) 2 to 3 inches below the bottom edge of combination label.	301	LSb-1	6-8	B	B
	c. Apply the size label (see 3.3.13.5) centered on the back side of the knee patches					
13.	Clean proximity trouser cover.					
	a. Trim all thread ends.					
	b. Remove all loose threads.					
	c. Remove all spots and stains.					
	NOTE: Care shall be taken in the use of solvents for cleaning to assure that the aluminized coating is not damaged.					

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3.8 Sizes and measurements. Sizes and measurements of the finished structural trouser and the proximity trouser cover shall be as shown in Tables V and VI (see Figures 1 and 2). All measurements and tolerances are expressed in inches.

TABLE V STRUCTURAL TROUSER

	(A)	(B)	(C) INSEAMS	
SIZES	WAIST	SEAT	REGULAR	LONG
34	35-1/2	48-1/2	26-1/2	31-1/2
36	37-1/2	50-1/2	27-1/2	32-1/2
38	39-1/2	52-1/2	27-1/2	32-1/2
40	41-1/2	54-1/2	27-1/2	32-1/2
42	43-1/2	56-1/2	27-1/2	32-1/2
44	45-1/2	58-1/2	27-1/2	32-1/2
46	47-1/2	60-1/2	28-1/2	33-1/2
Tolerance	±1/2	±1	±1/2	±1/2

TABLE VI - PROXIMITY TROUSER COVER

	(A)	(B)	(C) INSEAMS	
SIZES	WAIST	SEAT	REGULAR	LONG
34	36	49	28 1/2	33 1/2
36	38	51	29 1/2	34 1/2
38	40	53	29 1/2	34 1/2
40	42	55	29 1/2	34 1/2
42	44	57	29 1/2	34 1/2
44	46	59	29 1/2	34 1/2
46	48	61	30 1/2	35 1/2
Tolerance	±1/2	±1	±1/2	±1/2

NOTE: The structural trouser and proximity trouser cover shall be laid flat and smooth with the front snap fasteners and fastener tapes closed.

Waist (A) - Double the measurement taken straight across the front of the structural trousers or proximity cover 3/4 inch down from top of waistband from folded edge to folded edge.

Seat (B) - Double the measurement taken straight across the front of the structural trouser or proximity trouser cover from folded edge to folded edge, 1 inch below the bottom of fly.

Inseam (C) - The measurement shall be taken along the inseam from the junction of the front center seam to the bottom of the structural trouser or proximity trouser cover.

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3.9 Workmanship. The finished structural trouser and proximity trouser cover shall conform to the quality established by this specification. The occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows.

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

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4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be furnished for paragraph 3.3.3 - Moisture barrier and 3.3.4 - Cloth, abrasion resistant requirements. In addition, testing shall be performed on the components and characteristics listed in Table VII. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table VII shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

LOT SIZE	SAMPLE SIZE
800 or less	2
801 thru 22,000	3
22,001 and over	5

The basis for lot size and sample units for testing shall be as follows:

COMPONENT	LOT SIZE EXPRESSION	SAMPLE UNIT
Moisture barrier	yards	2 yards full width
Cloth, abrasion resistant	yards	2 yards full width

TABLE VII - TESTING OF COMPONENTS

COMPONENT	CHARACTERISTIC	REQUIREMENT PARAGRAPH	TEST METHOD/ PARAGRAPH
Moisture barrier	Coating compound	3.3.3	1/
	Weight	3.3.3	5041
	Breaking strength	3.3.3	5100
	Tear strength	3.3.3	5134
	Hydrostatic resistance	3.3.3	5512 2/
	Flame resistance	3.3.3	5903.1
Cloth, abrasion resistant	Base fabric	3.3.4	1/
	Type of weave	3.3.4	1/
	Coating compound	3.3.4	1/
	Weight	3.3.4	5041
	Thickness	3.3.4	5030
	Breaking strength	3.3.4	5100
	Tear strength	3.3.4	5134
	Flame resistance	3.3.4	5903.1
	Abrasion resistance	3.3.4	5306 3/

1/ A certificate of compliance shall be acceptable for this requirement.

2/ The water pressure shall be applied to the coated side of the material.

3/ Using the H18 abrasion wheel, 500 gram load on each abrasion arm for 2000 cycles.

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4.4.2 Examination of the end item structural trouser and proximity trouser cover. Examination of the end item shall be in accordance with MIL-STD-1666 for Material Defects and Damages (Sec.I), Components and Assembly (Sec.II), Cutting (Sec.III), Cleanness (Sec.IV), Seams and Stitchings (Sec.V), Metal Components (Sec.VII), Dimensional examination specified in paragraph 4.3, Inspection levels and acceptable quality levels specified in paragraph 4.4, and examination of packaging requirements specified in 4.5, also the defects listed in 4.4.2.1 of this specification. Defects listed in paragraph 4.4.2.1 shall take precedence over defects listed for similar examination in MIL-STD-1666.

4.4.2.1 Visual examination. The structural trouser and proximity trouser cover shall be examined for defects in construction, design and workmanship as follows:

NOTE: When an asterisk (*) appears, the defect shall be classified as follows:

MAJOR - When seriously affecting serviceability or appearance.
MINOR - When affecting serviceability or appearance, but not seriously.

CLASSIFICATION OF DEFECTS

DEFECT	CLASSIFICATION MAJOR	MINOR
REFLECTIVE TAPE		
a. One section missing		X
b. More than one section missing	X	
c. Edge of tape not parallel with bottom edge of trouser legs by more than 1/2 inch		X
FASTENER TAPE		
a. Any section missing	*	
b. Any section incorrect component, i.e. hook portion instead of loop portion or loop portion instead of hook portion	*	
c. Not edge stitched along all edges when specified		X
WEBBING		
a. Not tightly woven		X
b. Raveled edge		X
TOP EDGE OF TROUSERS		
a. Excessive fullness, twist or puckering at joining to waist facing		X
b. Waist facing exposed beyond top edge of trousers (score only when the condition exists on more than 1/2 the length of the seam)		X

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DEFECT	CLASSIFICATION MAJOR	MINOR
FLY CONSTRUCTION AND CLOSURE		
a. Waist edges not even by more than 3/8 inch when fly is fastened		X
b. Sockets and studs out of alignment causing a noticeable bulge or twist on front fly when fastened	X	
c. Fly base noticeably twisted or puckered		X
ABRASION RESISTANT CLOTH KNEE PATCHES (structural trouser only)		
a. Abrasion resistant cloth knee patch missing	X	
b. Abrasion resistant cloth knee patch misplaced or not attached as specified		X
LEG TABS (structural trouser only)		
a. Leg tab missing	X	
b. Leg tab less than 1 inch wide		X
c. Leg tab less than 3-3/4 or more than 4-1/4 inches long		X
d. Leg tab not attached or positioned as specified		X
HEM (structural trouser only)		
a. Width of hem less than 1-3/8 or more than 1-5/8 inches		X
b. Not uniform in width by more than 1/4 inch		X
LEG PROTECTORS (proximity trouser cover only)		
a. Leg protector missing	X	
b. Leg protector misplaced or not attached as specified	X	
c. Elastic tubing caught in stitching		X
LEG BOTTOMS (proximity trouser cover only)		
a. Bottom of legs not abrasion resistant cloth bound		X
b. Abrasion resistant cloth binding on legs overlapped less than 1 inch or more than 1-1/2 inch		X
KNEE PATCHES (proximity trouser cover only)		
a. Knee patch missing	X	
b. Knee patch misplaced or not as specified		X
c. Knee patch holder missing	X	
d. Knee patch holder misplaced or not as specified		X

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CLASSIFICATION OF DEFECTS

DEFECT	CLASSIFICATION MAJOR	MINOR
LABEL		
a. Omitted, one or more		X
b. Not attached as specified		X
c. Inscription or instruction omitted, incomplete, illegible or incorrect		X
d. Position not as specified		X
INSTRUCTION MANUAL		
a. Omitted or not positioned as specified		X

4.4.3 Examination of packaging requirements. An examination shall be made to determine that preservation, packing, and marking comply with section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

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4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. Packaging

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Structural trouser. Structural trouser shall be folded as follows:

1. Insert instruction manual into the cargo pocket.
2. Close front fly.
3. Place fastened suspenders inside structural trouser.
4. Fold structural trouser in half lengthwise with the fly closure on the outside of the fold.
5. Fold in half across the length by bringing the bottom of the structural trouser up to the top.

The folded structural trouser shall measure approximately 23 inches by 15 inches. Each folded structural trouser shall be banded with a commercially acceptable paper strip across the middle, with the ends secured with gummed paper tape.

5.1.1.2 Proximity trouser cover. Proximity trouser cover shall be folded as follows:

1. Close front fly.
2. Fold proximity trouser cover in half lengthwise with the fly closure on the outside of the fold.
3. Fold in half across the length by bringing the bottom of the proximity trouser cover up to the top.

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Folded proximity trouser cover shall measure approximately 23 inches by 15 inches. Each folded proximity trouser cover, shall include an instruction manual positioned so that it is visible, and sets of spare knee patches individually packaged in their own snug-fitting, clear polyethylene film bag conforming to A-A-50083. Each set of spare knee patches shall then be packed within the proximity trouser cover unit pack.

5.1.2 Commercial. Structural trouser and proximity trouser cover with spare knee patches included, shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Structural trousers. Five structural trousers, of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Inside dimensions of the box shall be approximately 23-1/2 inches in length, 15-1/2 inches in width and 15-1/2 inches in depth. and Each box shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636.

5.2.1.2 Proximity trouser covers. Five proximity trouser cover unit packs of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Inside dimensions of the box shall be approximately 23-1/2 inches in length, 15-1/2 inches in width and 15-1/2 inches in depth. Each box shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Structural trousers. Five structural trousers of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275, of PPP-B-636. Inside dimensions of the box shall be approximately 23-1/2 inches in length, 15-1/2 inches in width and 15-1/2 inches in depth. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.2 Proximity trouser covers. Five proximity trouser cover unit packs of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275, of PPP-B-636. Inside dimensions of the box shall be approximately 23-1/2 inches in length, 15-1/2 inches in width and 15-1/2 inches in depth. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.3 Commercial packaging. Structural trousers and proximity trouser covers with spare knee patches included, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry),type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with straps in accordance with bonding means C and D or Film bonding means F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall only indicate the following information:

STOCK NUMBER
NOMENCLATURE
SIZE
QUANTITY

5.5 MIL-STD-2073. When specified (see 6.2), Preservation and Packaging shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

6.1 Intended use. The trousers are intended for wear by DOD personnel engaged in firefighting. The structural trouser is for use by firefighters engaged in structural type fires. The structural trouser with the proximity trouser cover attached is for use by firefighters engaged in proximity type fire fighting.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Types, sizes and lengths required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
d. When first article inspection is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, testing and approval of the first article.
e. Levels of preservation and packing required (see 5.1 and 5.2).
f. When palletization is required (see 5.3).
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

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6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

6.4 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

6.5 A suggested source of supply for the moisture barrier lining material specified in 3.3.3 is Aldan Rubber Co., Tioga and Salmon Streets, Philadelphia, PA 19134

6.6 A suggested source of supply for the cloth, abrasion resistant material specified in 3.3.4 is Southern Mills, 6501 Mall Boulevard, P.O. Box 289, Union City, GA 30291

6.7 A suggested source of supply for the para-aramid twill material specified in 3.3.5 is Southern Mills, 6501 Mall Boulevard, P.O. Box 289, Union City, GA 30291.

6.8 A suggested source of supply for the heat sealing tape specified in 3.3.7 is Electro Seal Corp., 55 West Anaque Ave., Pompton, NJ 07442.

6.9 A suggested source of supply for the reflective tape specified in 3.3.8 is 3M Safety and Security Systems Division, St. Paul, MN 55144.

6.10 A suggested source of supply for the tubing connectors specified in 3.3.10.2 is McMaster Carr Supply Co., 600-T County Line Road, Elmhurst, IL 60126.

6.11 A suggested source of supply for the suspenders specified in 3.3.11 is Body Guard (Division of Lion Apparel) P.O.Box 14576, Dayton, OH 45413-0576.

6.12 A suggested source of supply for the combination stud and socket specified in 3.3.12.2 is Scovill Inc., 1100-T Buckingham St., Watertown, CT 06795.

6.13 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.15 Subject term (key word) listing.
Cloth, para-aramid/PBI
Clothing, protective
Coat, firefighter's
Liner, quilted
Suit, firefighters, combination

Custodians:
Navy - NU
Air Force - 99

Preparing activity:
DLA-CT

Project No. 8415-0929

Review activities:
Navy - MC
Air Force - 82,50,45

User activities:
Navy - SH
Coast Guard-CG

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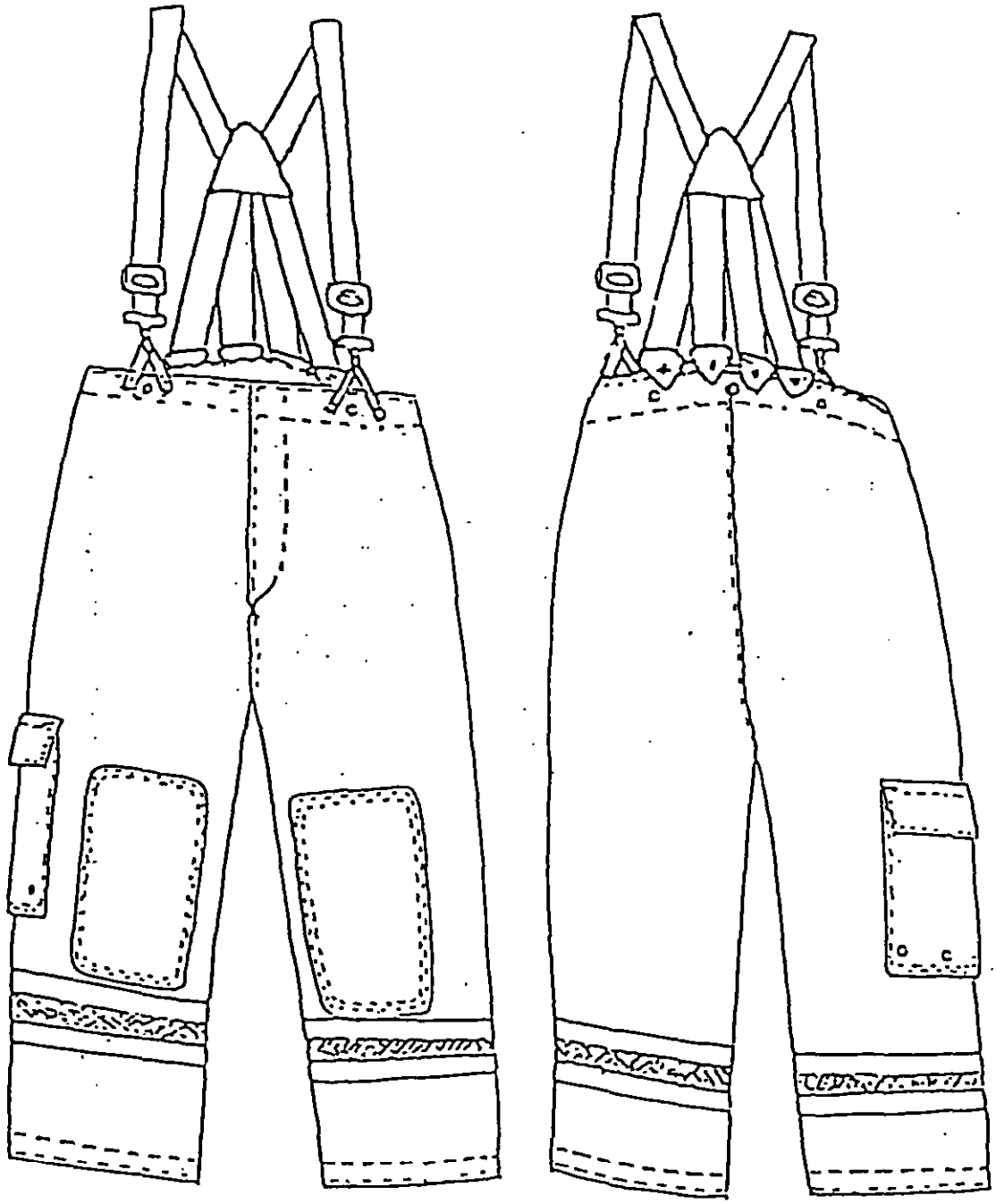
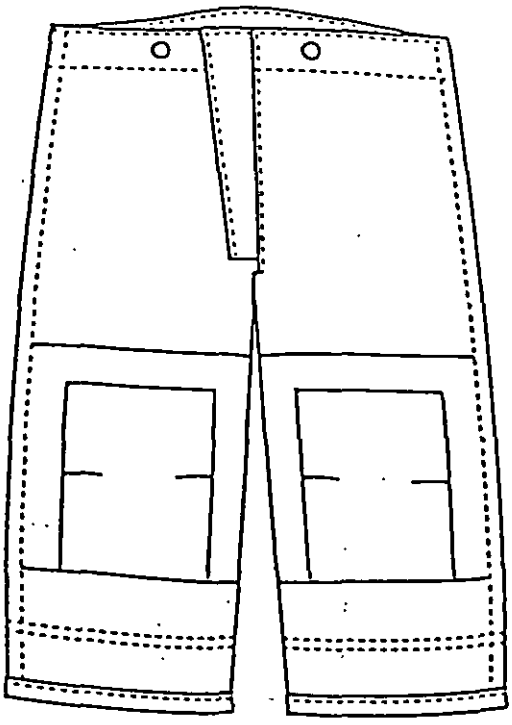


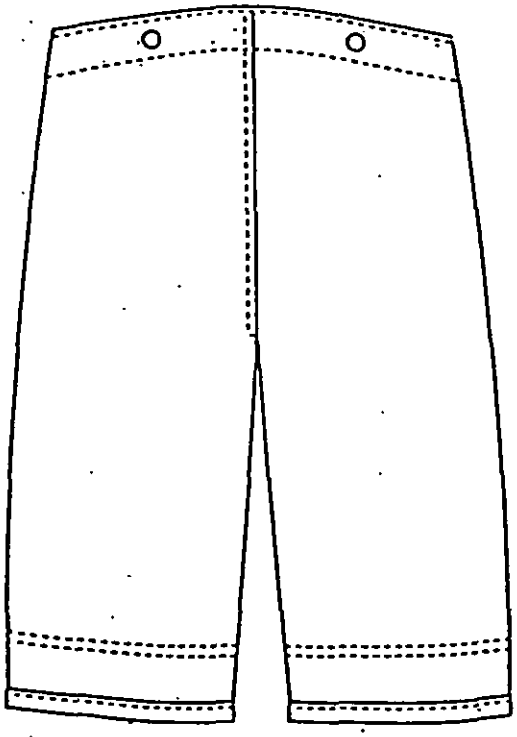
FIGURE - 1

STRUCTURAL TROUSERS

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FRONT VIEW



BACK VIEW

FIGURE 2

COVER, FIREMEN'S, TROUSERS (PROXIMITY)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-24940A	930811

3. DOCUMENT TITLE
TROUSER, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Personnel Support Center ATTN: DPSC-FSSD (12-3-D)	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) 2800 South 20th Street P. O. Box 8419 Philadelphia, Pennsylvania 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89

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